

Modernizing Data Warehousing for BI Analytics



Introduction

The R3 Research process, a cornerstone of business operations, originally relied on a complex system of Stored Procedures in SQL Server to extract data from Oracle databases. Leveraging a robust technology stack a comprehensive modernization initiative was undertaken to address challenges such as intricate data staging, difficulty in code enhancements, and a lack of effective monitoring and maintenance.

Business Challenge

The R3 Research process is vital for the business, initially relying on Stored Procedures in SQL Server to extract data from Oracle into SQL Server, and then into Google Big Query.

However, this approach faced below challenges:

- ✦ Complex data staging before loading into Google Big Query.
- ✦ Difficulty in enhancing existing code.
- ✦ Lack of monitoring and maintenance for the SQL Server-based process.

Technology Stack



◎ Solution And Highlights

The revamped R3 Research process now utilizes Python scripts to convert stored procedures into an ETL Process, executed via Informatica workflows six times daily.

Key highlights include:

- ✦ Direct sourcing of data from Oracle into Google Big Query, eliminating the need for an intermediary SQL Server data source.
- ✦ Quick issue remediation in case of failures.
- ✦ Seamless addition of code enhancements.
- ✦ Enhanced performance, maintenance, and availability.

◎ Benefits And Results

The modernized R3 Research process yields the following benefits:

- ✦ Efficient and trouble-free Looker dashboards and reports built on the Google Big Query R3 Research table.
- ✦ Active maintenance team involvement ensures prompt issue resolution.
- ✦ Improved process performance and data availability in Google Big Query, surpassing the previous method.

◎ Conclusion

The transformation of the R3 Research process has substantially improved the efficiency, reliability, and maintainability of data operations. The enhanced performance and availability of data led to better business decision-making.